QQ-A-250/9F
January 18, 1971
SUPERSEDING
Fed. Spec. QQ-A-250/9E
March 17, 1967

FEDERAL SPECIFICATION

ALUMINUM ALLOY 5456, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250.)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 5456 plate and sheet; the general requirements are covered in QQ-A-250.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified in one of the following tempers, as specified (see 6.2):

O - Annealed.
H321 - Strain-hardened one-quarter hard (hot rolled) temper and then stabilized.
H323 - Strain-hardened one-quarter hard (cold rolled) temper and then stabilized.
H343 - Strain-hardened half-hard (cold rolled) temper and then stabilized.
H112 - As rolled, properties as specified in table II.

2. APPLICABLE DOCUMENTS

Latest issue of Federal Specification:

QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.

FSC 9535

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3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition1/

Element	Percent	
	Minimum	Maximum
Magnesium	4.7	5.5
Manganese	0.50	1.0
Chromium	0.05	0.20
Iron plus silicon	-	.40
Zinc	-	.25
Titanium	-	.20
Copper	-	.10
Others, each	-	.05
Others, total	-	.15
Aluminum	Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties.

3.2.1 Mechanical properties of material as supplied. The mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

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TABLE II. Mechanical properties

Temper	Thickness	Tensile strength		Yield strength 1/		Elongation in 2 in. or 4 times D 2/, 3/, minimum
		Minimum	Maximum	Minimum	Maximum	
	Inches	P.s.i.	P.s.i.	P.s.i.	P.s.i.	Percent
O	0.051 thru 1.500	42,000	53,000	19,000	30,000	16
	1.501 thru 3.000	41,000	52,000	18,000	30,000	16
	3.001 thru 5.000	40,000	-	17,000	-	14
	5.001 thru 7.000	39,000	-	16,000	-	14
	7.001 thru 8.000	38,000	-	15,000	-	12
H112	0.250 thru 1.500	42,000	-	19,000	-	12
	1.501 thru 3.000	41,000	-	18,000	-	12
H321	0.188 thru 0.624	46,000	59,000	33,000	46,000	12
	.625 thru 1.250	46,000	56,000	33,000	45,000	12
	1.251 thru 1.500	44,000	56,000	31,000	43,000	12
	1.501 thru 3.000	41,000	56,000	29,000	43,000	12
H323	0.051 thru 0.125	48,000	58,000	36,000	46,000	6
	.126 thru ..249	48,000	58,000	36,000	46,000	8
H343	0.051 thru 0.125	53,000	63,000	41,000	51,000	6
	.126 thru .249	53,000	63,000	41,000	51,000	8

1/ At 0.2 percent offset.

2/ Not required for material 1/2 inch or less in width.

3/ D represents specimen diameter.

4. QUALITY ASSURANCE PROVISIONS
(See QQ-A-250.)

5. PREPARATION FOR DELIVERY
(See QQ-A-250.)

6. NOTES

6.1 Intended use. Sheet and plate of aluminum alloy 5456 are used for applications requiring weldable, moderate strength, nonheat-treatable alloy.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Form and temper required (see 1.2.1).

(c) Dimensions required.

(d) Requirements for sizes not specifically covered (see QQ-A-250).

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(e) Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

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SPECIFICATION

QQ-A-250/9F ALUMINUM ALLOY 5456, PLATE AND SHEET

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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